AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 15 · TEACHER GUIDE

Save New Examples for Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach independent evaluation as an ordered protocol.

Distinguish public worked examples, supported RodD practice and the separate independent RibbonA experiment.

Keep one habit: Look for saved pre-label predictions, exact versions and honest limits.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Separate examples used to choose a model from examples reserved for its final evaluation

Make the order of **evidence** visible. The lesson table is a retrospective worked example, not an unseen student test. W4 is a separate staged exercise: do not reveal its labels before **prediction**s are committed.

Avoid calling ordinary **input** validation a validation set here; the focus is development versus a held-out final check.

By the end, I can **freeze** a model before labels are revealed and report a bounded test result.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Development and reserved tower cards from the worked example, marked as a completed experiment
- RibbonA **input** cards 5, 7, 9, 11 cm and a covered label sheet: short, short, long, long
- Paper for a dated frozen-rule record and separate **prediction**s, labels and outcome columns

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Construct a classifier with a stated **input** format, selected rule and recorded **output**

Reference example, not the only acceptable wording: Require a present height, unit cm, a whole number and range 1 through 10 inclusive.

For example, 7 cm is accepted; blank is missing, 60 mm has the wrong unit and 11 cm is outside the range. Fractional values such as 4.5 cm also fail.

The contract defines accepted **input**s; it does not list only the **fitting** records or establish **prediction** correctness.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Compare expected and observed traces to locate the first difference

Reference example, not the only acceptable wording: Expected trace: 3, 5, 10, 9. First difference is the doubling checkpoint: 10 expected versus 8 observed.

The later 8 minus 1 equals 7, so the observed subtraction is consistent with its **input**. Investigate the earlier step before claiming another fault.

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an ordered algorithm using the current value after every instruction

Reference example, not the only acceptable wording: The trace is 3, 7, 14, 9. The doubling step uses 7, the current value after adding 4 to 3. It must not double the initial 3.

Accept a table, arrows or clearly labelled sentences that preserve the order.

Compare the **prediction** with an appropriate recorded label.

If the label helps choose the threshold, that example has participated in **fitting** or model selection; it is not independent evaluation **evidence** for the resulting model.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Build a Human Classifier

Debug the Cause

Algorithms as Recipes

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Concept to explain

A fair test uses examples not used to create the decision **process**.

Term	Meaning
training set	examples used to create or adjust a process
test set	untouched examples used for final checking
holdout	data deliberately kept separate
data leakage	test information influences design

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the evidence protocol before reporting a score

> Worked labels are public practice; keep the distinct independent RibbonA labels private until its predictions are recorded.

Begin with the reserve-and-freeze protocol

A rule chosen to fit examples may work differently on new cases. To investigate that difference, reserve a set of examples before development. Use the **development examples** to choose the model.

Keep the final test examples separate until the model is fixed, then evaluate that recorded model.

In this paper activity, a teacher keeps the reserved examples in an envelope. Record the **input** contract, feature, category rule and model version before opening it.

Once the **input** values are revealed, predict using the frozen model before revealing their comparison labels. This prevents copying the labels into the **prediction**s.

The worked example below describes a completed experiment, so its test labels are visible for discussion. It is not an unseen test for you.

In the workbook experiment, the teacher will supply a separate set of comparison labels after you have saved your **prediction**s. All data are imaginary classroom records.

Identify the completed TowerA experiment

In the completed experiment, development towers at heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Candidates predict tall at or above 5 or 6 cm.

Both fit all four labels; the stated smaller-threshold tie rule selects 5. The team **freeze**s model TowerA with threshold 5.

The reserved **input**s are different records at heights 3, 5, 7 and 9 cm. TowerA accepts whole cm heights from 1 through 10. It predicts short, tall, tall and tall.

These **prediction**s are saved before the reserved labels are opened. The comparison below records what happened, including the failure.

Model the completed PencilP evaluation

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

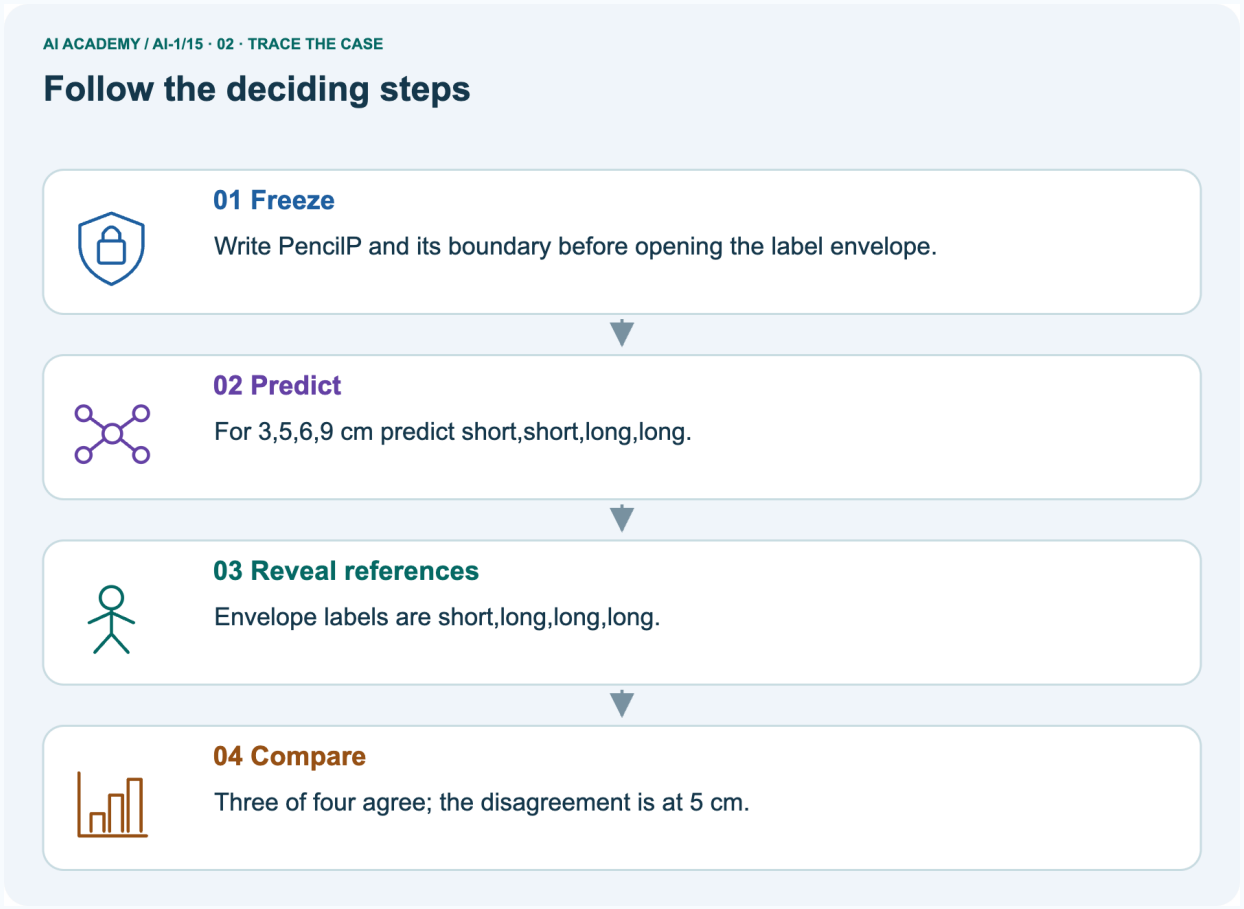
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Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Revise PencilP after its visible feedback

1. Change: After feedback, change the boundary to 5.
2. **Prediction**: The revised rule matches all four known labels but needs a different final test.
3. Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Limit conclusions beyond the selected cases

Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Preserve TowerA's result beside the TowerB replay

Reserved height	Saved TowerA prediction	Revealed comparison label
3 cm	Short	Short
5 cm	Tall	Short
7 cm	Tall	Tall
9 cm	Tall	Tall

TowerA agrees on three of these four reserved examples and disagrees at 5 cm. That is **evidence** about this fixed model on this small set. It does not prove a success rate for every future tower.

The labels, range of examples and way records were selected also matter.

After seeing the 5 cm label, the team changes the threshold to 6 and names the revision TowerB. TowerB agrees on all four of these now-seen examples.

The change may be useful, but those examples have influenced its development. Four agreements on them are not an independent final test result for TowerB. Keep TowerA's original three-of-four record.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** all labeled cards

> **2. Process — the deciding step**
>
> **PROCESS** separate **training** and untouched testing before rule creation

> **3. Output — the result**
>
> **OUTPUT** honest performance **evidence** on new cards

CORE IDEA A fair test uses examples not used to create the decision **process**.

UNDER THE HOOD Evaluation on **training** data is optimistically biased. A **holdout test set** estimates behavior on unseen examples only if it remains untouched during design.

Repeatedly tuning to test results leaks information and turns the test into **training**.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

STEP 1 — OBSERVE all labeled cards

STEP 2 — TRACE separate **training** and untouched testing before rule creation

STEP 3 — CONCLUDE honest performance **evidence** on new cards

MODEL REASONING Randomly or deliberately reserve representative cards first. Create the decision **process** using only the learning pile.

Open the test pile once for **evidence**, record every result, and do not call the revised **process** tested until a new **holdout** exists.

DO NOT OVERCLAIM A **test set** stops being independent when its answers guide revisions.

What new **evidence** could change this answer?

Concrete teaching case

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

Exact explanation to model

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Explain the added worked example

Key terms

Term	Meaning
Development examples	Examples available for fitting, comparing or revising a model
Held-out test set	Examples reserved from development for evaluating a fixed model
Freeze	Record a model and stop changing it for a particular evaluation
Test leakage	Information from a reserved test influencing model development
Evaluation record	A record of the tested model, predictions, comparison labels and results

Model the reasoning aloud

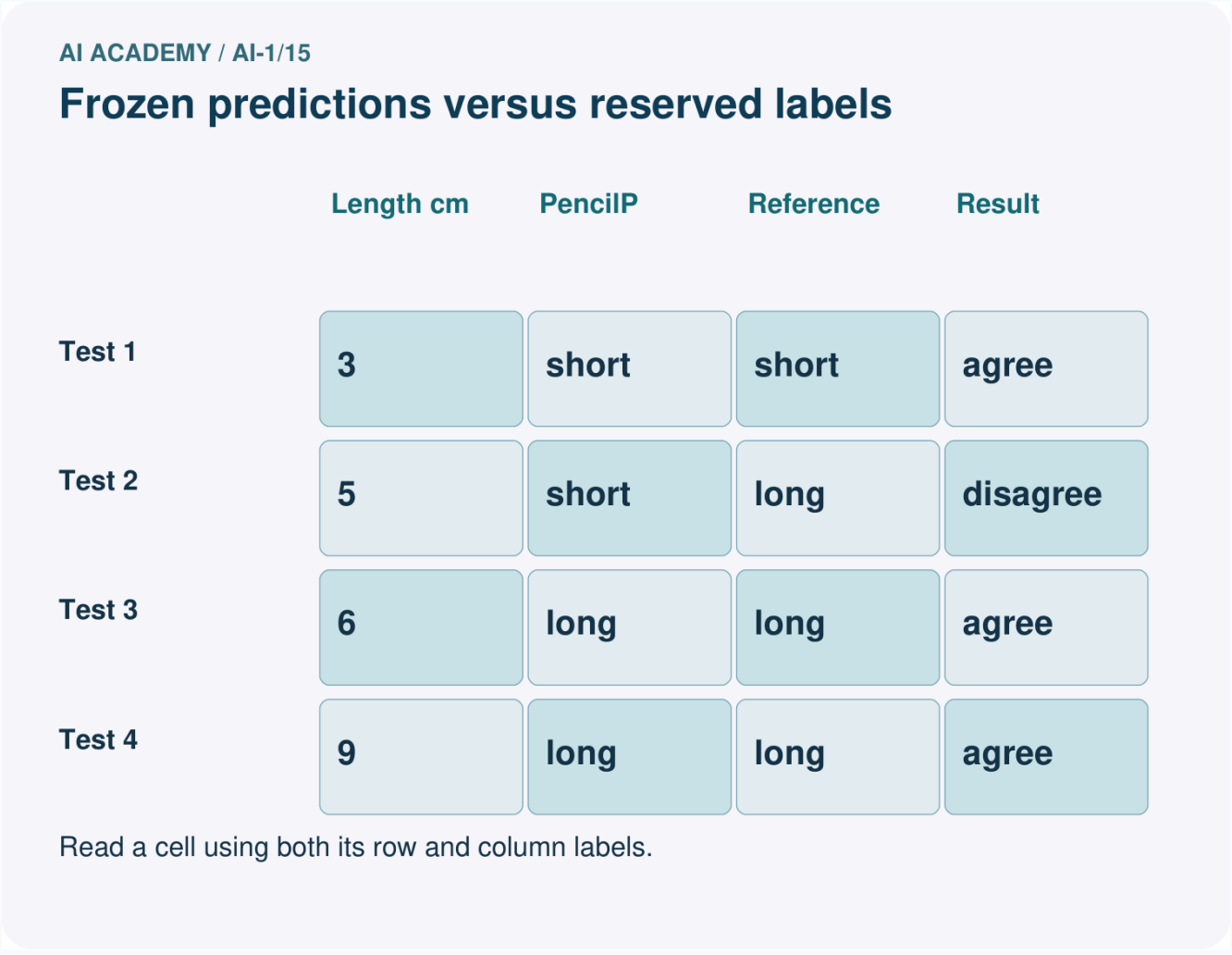
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Save New Examples for Testing

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

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Save New Examples for Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze

Write PencilP and its boundary before opening the label envelope.



2. Predict

For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references

Envelope labels are short,long,long,long.



4. Compare

Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING

After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze	Write PencilP and its boundary before opening the label envelope.
2. Predict	For 3,5,6,9 cm predict short,short,long,long.
3. Reveal references	Envelope labels are short,long,long,long.
4. Compare	Three of four agree; the disagreement is at 5 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its **prediction**s, agreements and disagreement.

Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

Expected reasoning and fading help

Teacher check: TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection.

Report its four agreements as a result on examples used in development, preserving TowerA's original test record.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Keeping final tests unseen. Proposed explanation: "A new file makes old records unseen". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Trace whether records or copies already influenced development, regardless of their file names.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Does 4/4 on the replay replace the original 3/4 result? Explain.

No. Keep 3/4 as the frozen version's result; record 4/4 separately as a revised-version replay on known cases.

Changed-case reasoning: The revised rule matches all four known labels but needs a different final test. These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Cards A–F are used for practice. G and H remain sealed.

Write the question to explain: Which can provide a fresh test, and under what condition?

Reveal after **prediction**s: G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Use the earlier changed question as optional later practice. Reserve Run a separate ribbon experiment for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

TowerA agreed on three of four original reserved records, failing at 5 cm. TowerB agreed on all four after their labels informed its revision; this is performance on development **evidence**.

Freeze TowerB before evaluating fresh appropriate reserved examples. Shuffling cannot undo prior use.

Exit: the team selected using test scores, so it measured a development comparison and still lacks an independent final check of the winner.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the difference between accepted input and checked prediction. Ask when a label becomes part of development.
7 to 20 minutes	Trace the three evidence stages in the diagram. Explicitly identify the lesson table as an already completed experiment.
20 to 33 minutes	Compare TowerA predictions and revealed labels. Complete W1 and W2, preserving the original failure when TowerB is introduced.
33 to 45 minutes	Use W3 to show how a score can influence model selection without exposing individual labels. Discuss why shuffling is not a repair.

Stage	Teacher action
Session 2 start 0 to 8 minutes	Retrieve the frozen-model and saved-prediction requirements. Keep the RibbonA label sheet covered.
8 to 23 minutes	Complete the guided task and plan the two statements for TowerA and TowerB. Describe what fresh evidence would be needed.
23 to 38 minutes	Collect W4 predictions independently before revealing short, short, long, long. Students append results without erasing predictions, then explain the revised threshold.
38 to 45 minutes	Use W5 and the exit to distinguish independence from coverage. Keep original and revised result claims side by side.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Prepare RodD's separate supported-exercise envelope labels as SHORT,LONG,LONG,LONG for reserved inputs 2,3,4,6 cm. Do not reveal them before predictions are saved. Both candidates fit the two development records; the larger tie rule selects threshold 4. Its saved predictions are SHORT,SHORT,LONG,LONG: 3/4 agree, with a mismatch at 3. A feedback-

guided threshold 3 matches the now-seen records, but that replay is development evidence for the revision. Preserve the original RodD result. These are not the independent RibbonA inputs or labels; reserve Run a separate ribbon experiment for that first attempt.

Run and record

1. Complete the task: Split card data into learn and test piles before sorting.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A new file makes old records unseen: Trace whether records or copies already influenced development, regardless of their file names.

Only labels copied into **training** can leak: A test score can guide rule selection even when individual labels remain hidden.

Testing means the model can never improve: Improve openly, preserve the original result, and obtain fresh independent **evidence** for the revision.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Run a separate ribbon experiment for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR A card remains unseen after it has been used for teaching.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A perfect reserved score proves universal correctness: Check sample size, variety, relation to development records and intended use; keep the conclusion within the **evidence**.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

Saved **prediction**s are short, long, long, long. Reveal labels only after **prediction**s are saved: short, short, long, long. RibbonA has three agreements and one disagreement at 7 on four reserved records.

Threshold 8 then gives short, short, long, long, but it was chosen after feedback. Its four agreements are on now-seen **development examples**, not a new independent test. Do not alter the saved RibbonA **prediction**s.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Assign the evidence its role

For each event, state whether examples are being used for development or independent evaluation of a fixed model: comparing candidates on **fitting** labels; scoring a frozen model on previously reserved examples once; changing a threshold after reading a test score.

Explain your last choice.

Comparing candidates is development. Scoring a frozen model on previously reserved examples is evaluation, provided they did not influence its development.

Changing a threshold after reading the test score uses that test information for development of the revision; the same set is no longer independent of that revision.

W2 Preserve the original result

TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its **prediction**s, agreements and disagreement.

Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection.

Report its four agreements as a result on examples used in development, preserving TowerA's original test record.

W3 Detect a less visible leak

A team leaves reserved labels out of its **fitting** table but tries six rules and reads a test score for each. It chooses the highest-scoring rule. Explain how information still reached model selection.

Would renaming or reordering the same test records repair the problem?

The scores depend on the reserved labels and guide the choice among rules, so test information influences selection indirectly. Omitting the labels from a **fitting** table is insufficient.

Renaming or reordering records does not remove that information; reserve new appropriate examples for an independent final check.

W5 Audit a proposed final set

A class reserves ten copies of one easy development example in a new file. The model agrees with all ten labels.

Explain two reasons this gives weak **evidence** about new cases, and propose a better final evaluation plan without promising perfect future performance.

Copies repeat a case already used in development and provide almost no variety; a new file does not make them new **evidence**.

Reserve distinct relevant cases before model choices, keep related copies together, **freeze** the model, save **prediction**s, reveal labels and report every outcome.

Include cases relevant to intended use and state that a finite test cannot guarantee all future results.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Additional transfer practice question

You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

Reasoned practice answer

D and E provide new **evidence**. If D is shown during teaching, it is no longer an unseen test example; keep E or obtain another fresh card for testing.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A card remains unseen after it has been used for teaching.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. They became development **evidence** when they influenced revision. Reserve new independently labelled cases for final evaluation.

If none are available, disclose that limitation and report results as development results, not unseen performance.

Explanations for the additional workbook examples

W1 Assign the evidence its role

W2 Preserve the original result

W3 Detect a less visible leak

W4 Run a separate ribbon experiment

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

Saved **prediction**s are short, long, long, long. Reveal labels only after **prediction**s are saved: short, short, long, long. RibbonA has three agreements and one disagreement at 7 on four reserved records.

Threshold 8 then gives short, short, long, long, but it was chosen after feedback. Its four agreements are on now-seen **development examples**, not a new independent test. Do not alter the saved RibbonA **prediction**s.

W5 Audit a proposed final set

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use two envelopes and a timeline with a clear **freeze** point. Read the steps aloud, preserving the learner's original **prediction**s.

Extension: explain why selecting the best of many test scores may give a flattering result even without deliberate cheating.

W4 is secure with saved **prediction**s, three agreements, the 7 cm failure, a limited claim and a correct account of post-test revision. If labels were seen early, record the exercise as coached practice.

For an uncoached retry **freeze** BandA at long at least 4, use **input**s 2, 4, 6, 8, then reveal short, short, long, long: three agreements, failure at 4.

Plan the delayed check

Suggested return points in this level: ai-1/16, ai-1/18, ai-1/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 16: Boundaries and 'Not Sure'. Retrieve today’s **evidence** and identify what carries forward and what is new.

Week 16 develops explicit uncertainty and review paths when available **evidence** does not justify a confident category or action.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Separate examples used to choose a model from examples reserved for its final evaluation

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: TowerA uses threshold 5 on reserved heights 3, 5, 7 and 9 cm. The labels are short, short, tall and tall. Write its **prediction**s, agreements and disagreement.

Then state why TowerB with threshold 6, selected after seeing these labels, cannot claim four independent test successes on these same records.

Reasoned practice answer: TowerA predicts short, tall, tall, tall: three agreements and one disagreement at 5. TowerB predicts short, short, tall, tall, matching all four, but the revealed labels influenced its selection.

Report its four agreements as a result on examples used in development, preserving TowerA's original test record.

Check every learner before discussion: Use the worked case for Keeping final tests unseen. Proposed explanation: "A new file makes old records unseen". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Trace whether records or copies already influenced development, regardless of their file names.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: After feedback, change the boundary to 5. Require a **prediction** and a reason before comparing. Expected result: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ Reference: feedback and records

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

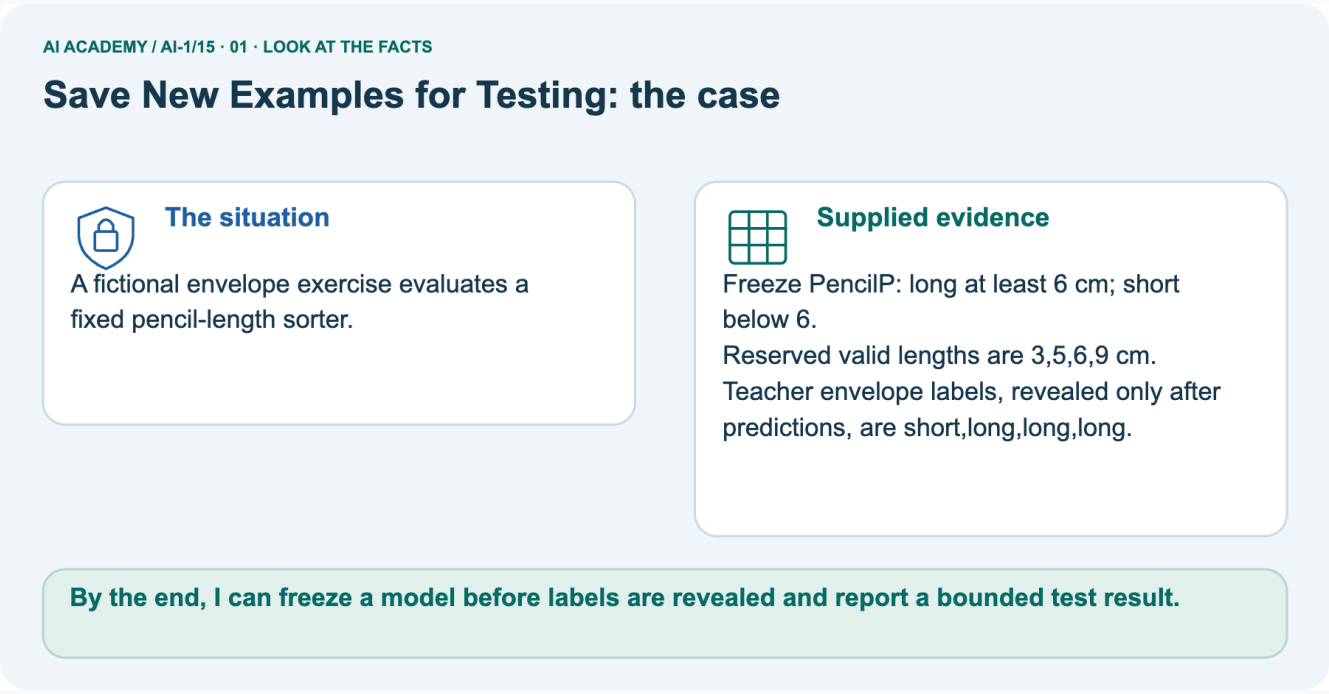
Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

▼ Reference: teach with the six pictures

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Save New Examples for Testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

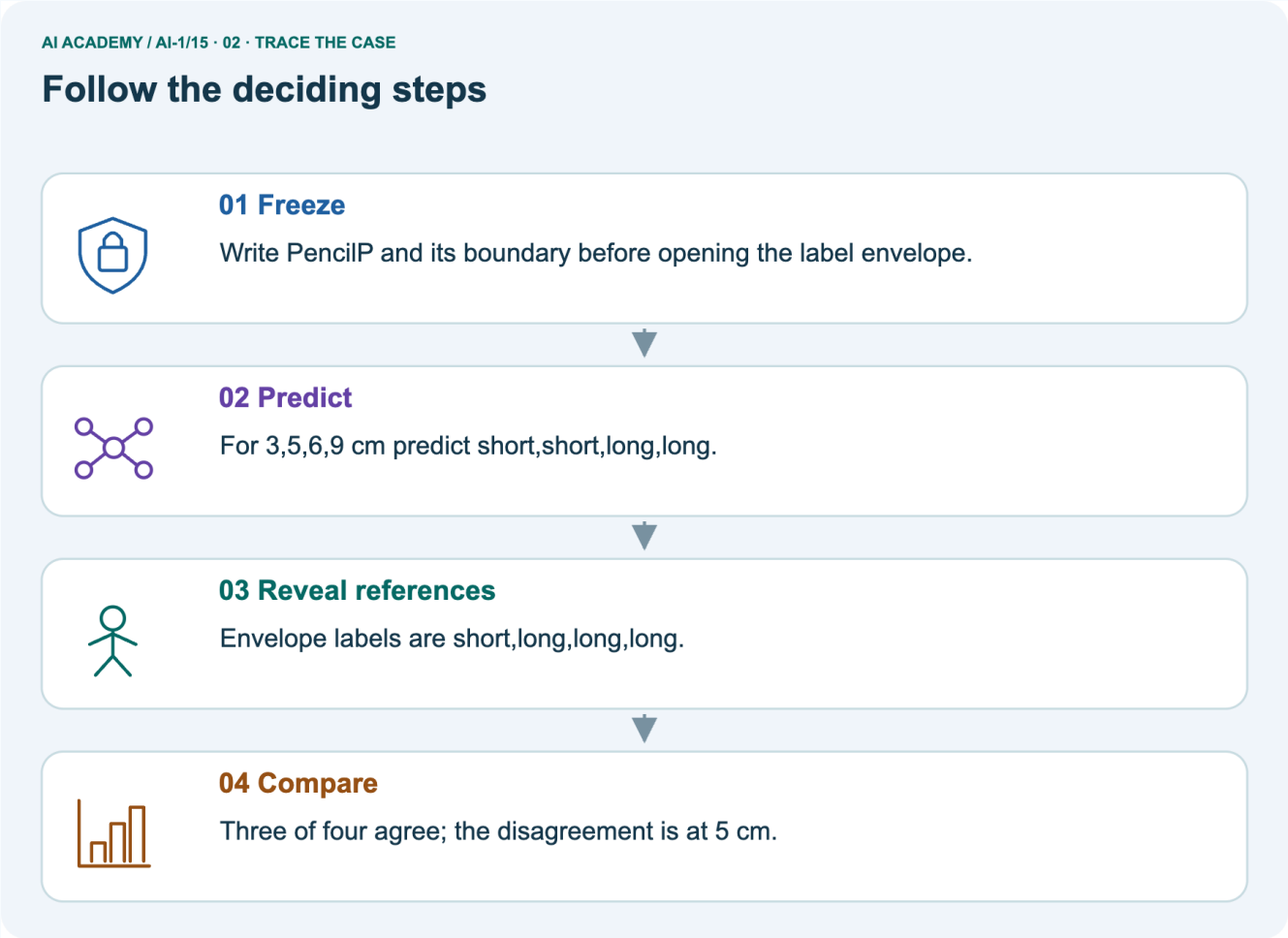
A fictional envelope exercise evaluates a fixed pencil-length sorter. Supplied facts: **Freeze** PencilP: long at least 6 cm; short below 6. Reserved valid lengths are 3,5,6,9 cm.

Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



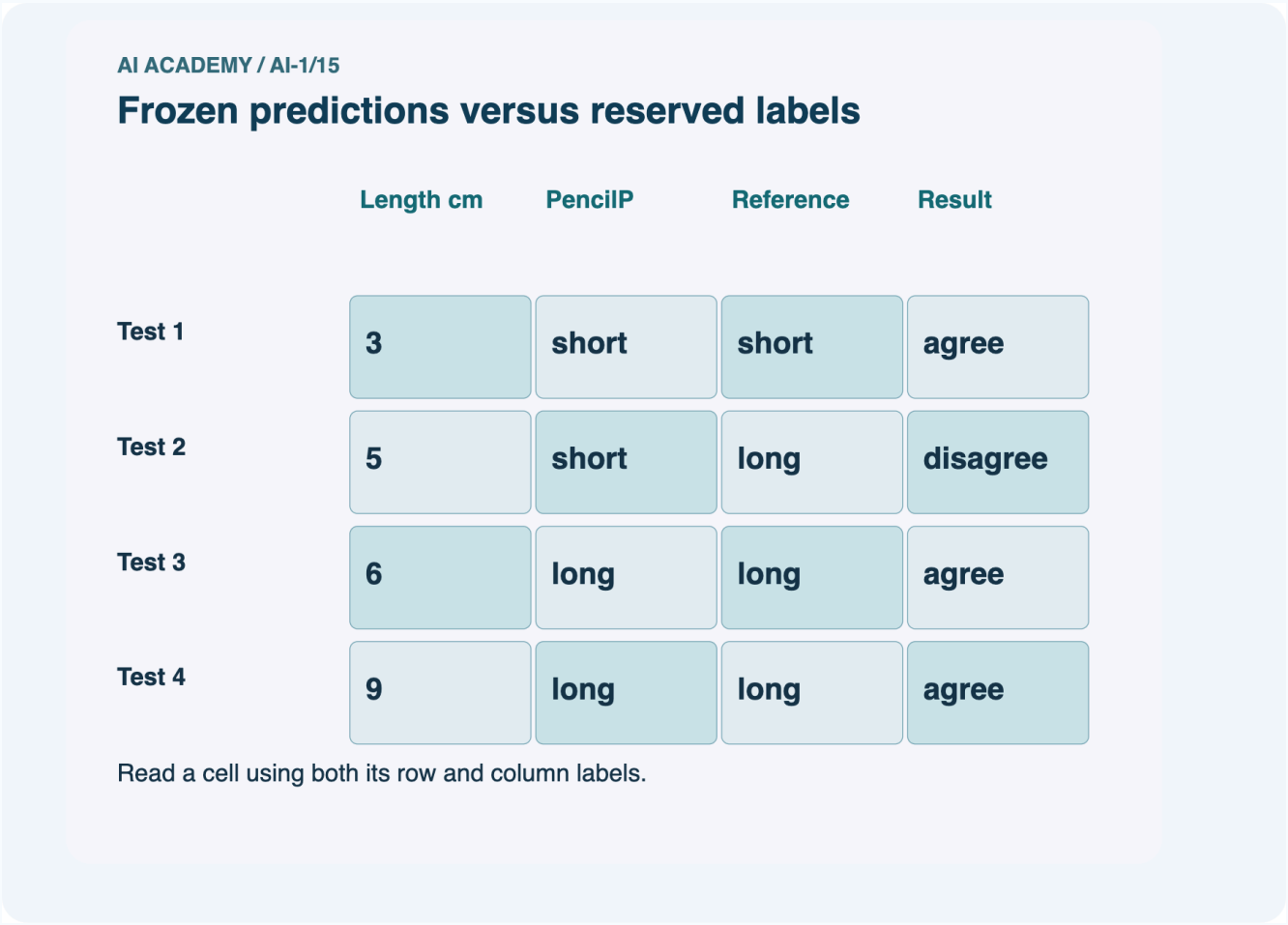
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze: Write PencilP and its boundary before opening the label envelope. → Predict: For 3,5,6,9 cm predict short,short,long,long. → Reveal references: Envelope labels are short,long,long,long. → Compare: Three of four agree; the disagreement is at 5 cm.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Frozen predictions versus reserved labels



Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen **prediction**s versus reserved labels. Test 1: Length cm = 3, PencilP = short, Reference = short, Result = agree. Test 2: Length cm = 5, PencilP = short, Reference = long, Result = disagree.

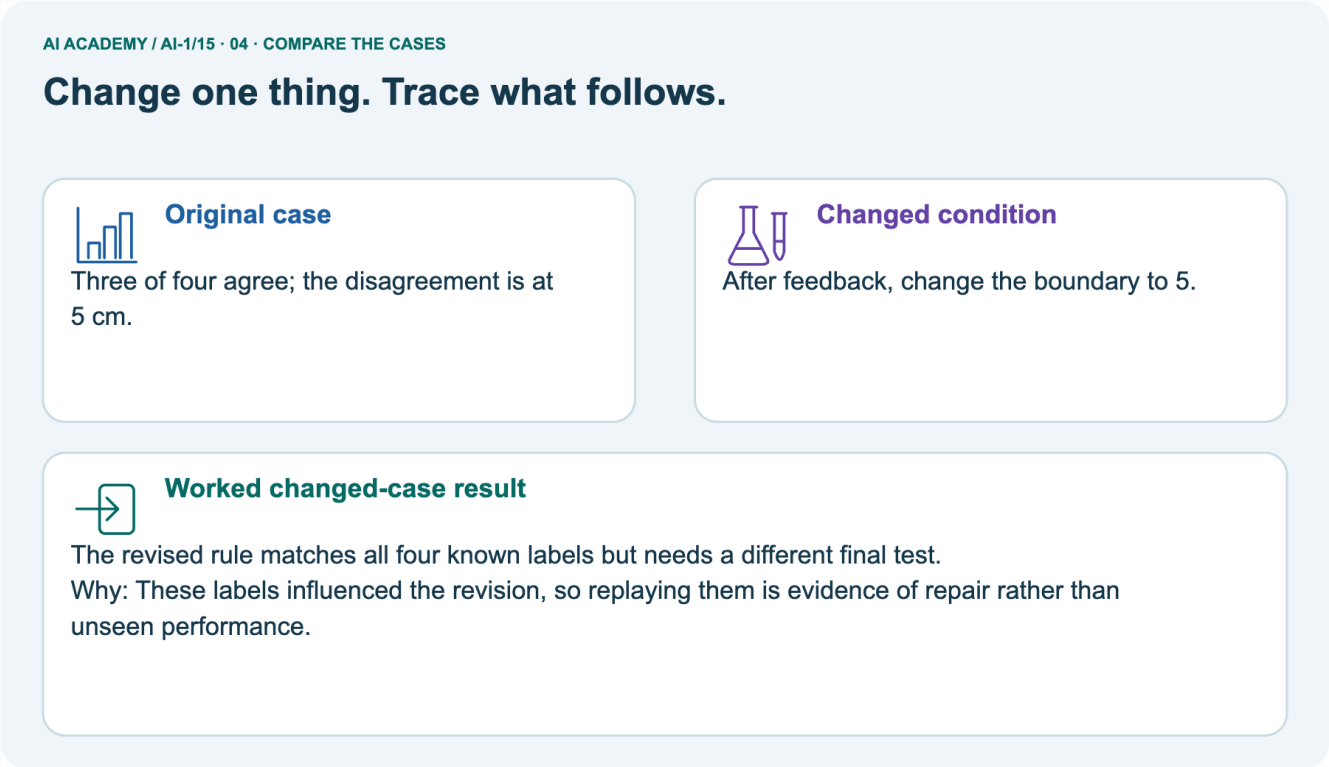
Test 3: Length cm = 6, PencilP = long, Reference = long, Result = agree. Test 4: Length cm = 9, PencilP = long, Reference = long, Result = agree.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four agree; the disagreement is at 5 cm. Changed condition: After feedback, change the boundary to 5. Expected result: The revised rule matches all four known labels but needs a different final test.

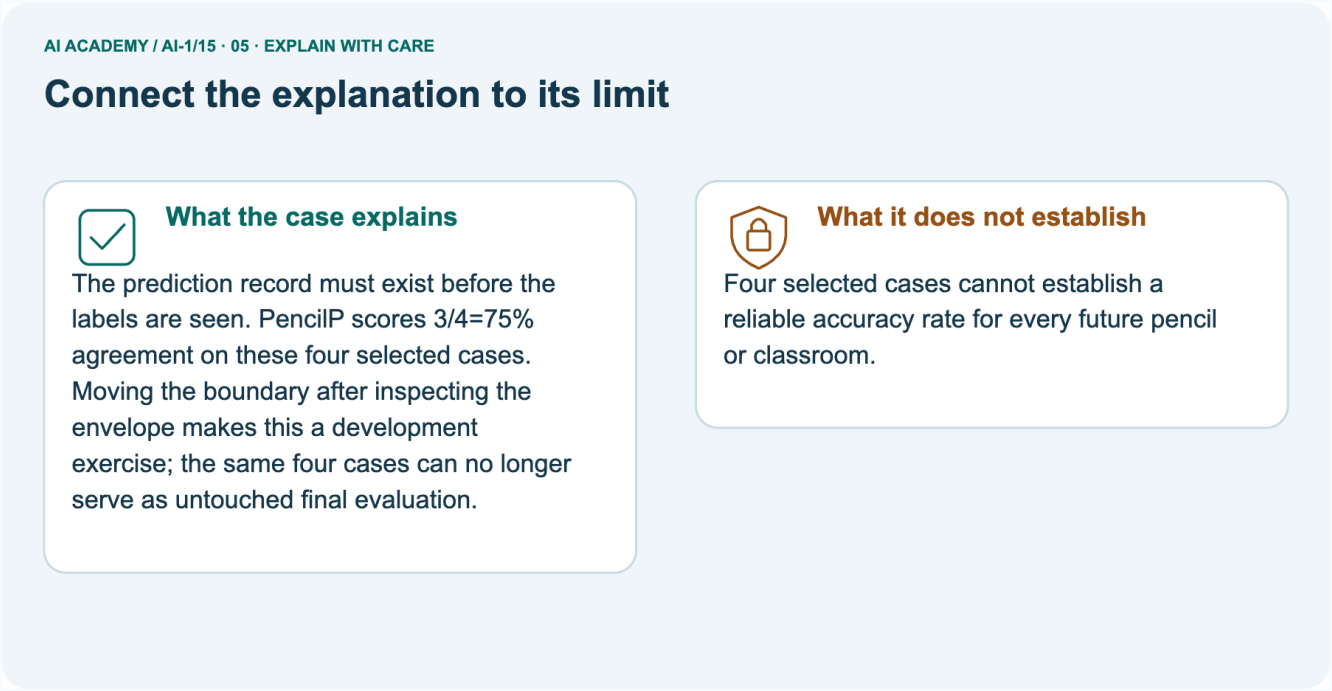
Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

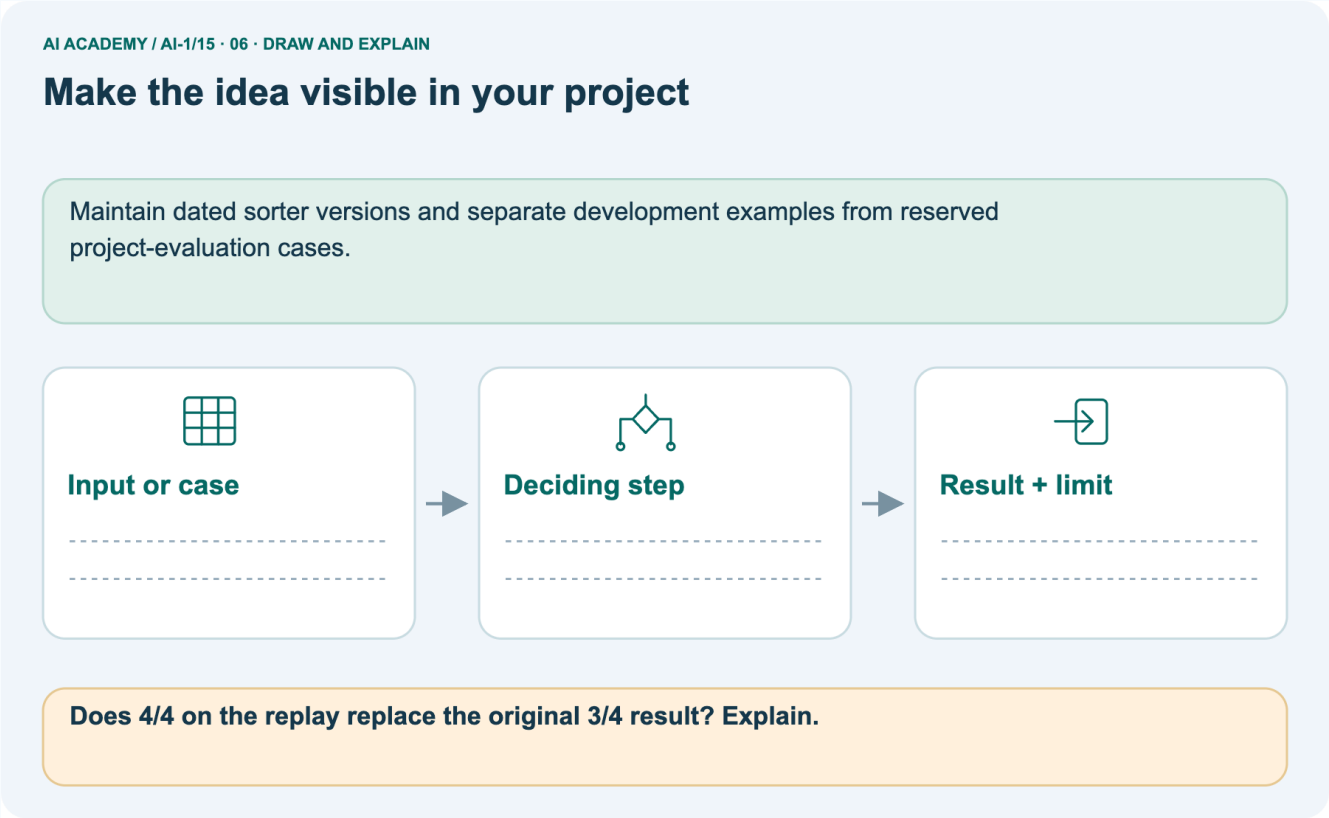
Limit: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases. Draw your **input**, deciding step, result and limit.

Supported question: Does 4/4 on the replay replace the original 3/4 result? Explain.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Keep 3/4 as the frozen version’s result; record 4/4 separately as a revised-version replay on known cases.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Save New Examples for Testing: the case

02. Follow the deciding steps

03. Frozen predictions versus reserved labels

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.